GRADE 2 · BLOCK CODING

Do They Meet?

Same number means they meet

Name: ....

Date: ....

★ **I can run two scripts at the same time and tell if the sprites land on the same number.**

Meeting on the path

Bit (green) and Pixel (purple) both start on the green flag ▶ and run at the SAME time. If they land on the SAME number, they MEET. If they land on different numbers, they do not. Read each script, find where each lands, then check: do they meet?

The number path

0	1	2	3	4	5
	Bit			Pixel	

Bit

Bit's script

▶ start at 1

forward 2

back 1

1 **Where does Bit land?**

Read Bit's script from 1: forward 2, then back 1. What number does Bit land on?

Pixel

Pixel's script

▶ start at 4

back 1

back 1

2 Where does Pixel land?

Read Pixel's script from 4: back 1, then back 1. What number does Pixel land on?

3 Do they meet?

The green flag starts BOTH scripts at once. Write where each lands. Bit: ____ Pixel: ____. Do they land on the SAME number? Circle YES or NO.

Big idea

Two scripts run at the same time. To tell if the sprites MEET, find where each one lands and check if the two numbers are the same. Same number = they meet.

TEACHER GUIDE · GRADE 2 · BLOCK CODING

Do They Meet?

Quick facilitation notes & answer key

What this worksheet teaches

Students run both characters' moves and decide: do they land on the SAME number (meet) or not? Holding both landings in mind to judge "meet" is the Grade 2 step. No coding experience needed.

Before you start (no computer needed)

No computer needed — paper and a pencil. You read the prompts aloud. Optional: put two counters on a number line 0 to 5 — green for Bit, purple for Pixel — and move BOTH at the same time so children see the two sets of moves run side by side.

How to lead it (about 20 minutes)

1. Show them (you do). Run both sets of moves and ask: "are they on the same number?"
2. Do one together. Exercise 1 — Bit lands on 2; Exercise 2 — Pixel lands on 2.
3. Let them try. Students write both landings and decide if they meet (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 — Bit: 1, forward 2 → 3, back 1 → 2. Exercise 2 — Pixel: 4, back 1 → 3, back 1 → 2.

Exercise 3 — both start on the green flag and both land on 2 → same number → YES, they meet.

Key idea: compare the two final positions; same number means they meet.

If students get stuck — and ways to adjust

Watch for: saying they meet because they are close. Meet means the exact same number.

Make it easier: move both counters and see if they land on the same spot.

Make it harder: ask what one move would make them NOT meet.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

In plain terms: students write and run instructions, including two sets that run at the same time.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

In plain terms: students read two sets of instructions, change one, and explain what happens when both run together.

You'll know it worked when

The child finds both landings and correctly decides whether they meet.